

SPMM
Smart
Revise

Liaison Psychiatry

Paper B

Syllabic content 7.3

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1. Specialist advice to medical specialities

A. Premenstrual Syndrome

PMS is a collection of psychological and somatic symptoms occurring during the luteal phase of menstrual cycle. Most of the woman (95%) have symptoms are of mild severity. However, 5% suffer from severe symptoms.

Severe PMS affects 3-8% women in reproductive age. Comorbidity with a mood disorder is seen in 30-70%. In those with PMDD, the risk of postnatal depression is higher.

ICD includes “premenstrual tension syndrome” under the heading “Diseases of the Genitourinary Tract”. Severe PMS was classified as premenstrual dysphoric disorder (PMDD) in DSM-IV under “depressive disorder not otherwise specified” along with a description attached to appendix B. With mounting evidence for its construct validity, it has been moved to the main text as a diagnosis in DSM-5. While many of its symptoms overlap with a depressive episode, the physical symptoms of breast pain and bloating are not seen in major depressive disorder.

Clinical criteria for PMDD. (Adapted from DSM-5): In most menstrual cycles during the past year, at least 5 of the following 11 symptoms (including at least 1 of the first 4 listed) should present:

- Depressed mood
- Marked anxiety
- Marked affective lability
- Marked anger or irritability (usually the most severe and start earlier)
- Anhedonia
- Subjective sense of difficulty in concentrating
- Lethargy, easy fatigability, or marked lack of energy
- Marked change in appetite, overeating, or specific food cravings
- Hypersomnia or insomnia
- A subjective sense of being overwhelmed or out of control
- Physical symptoms, e.g., breast tenderness or swelling, headaches, joint or muscle pain, a sensation of bloating, or weight gain

The symptoms must have been present for most of the time during the last week of the luteal phase, must have begun to remit within a few days of the onset of menstrual flow, and must be absent in the week after menses. Symptoms should markedly interfere with work, school, social activity or relationships. They cannot be just an exacerbation of another disorder such as depressive disorder. The criteria must be confirmed by prospective daily ratings for at least two consecutive menstrual cycles.

Pattern of symptom expression: During each cycle, PMS symptoms generally last for few days to 2 weeks. The peak is 2 days before the start of menses. Women tend to have the same pattern of symptoms in each cycle

Aetiology: Most accepted hypothesis- is increased sensitivity to normal, fluctuation of gonadal hormones. PMS is not caused due to drop in progesterone concentration. Serotonin has a role in the pathogenesis. This is supported by following evidence: serotonin-enhancing treatments reduces PMS symptoms, impairment in serotonin transmission provokes PMS symptoms and serotonergic transmission is aberrant in women with PMS. Imaging studies suggest GABA's role in PMS due interaction between progesterone metabolites and GABA-A receptors.

Treatment:

- Mild symptoms can be managed by making life style changes, CBT, exercise or dietary regulation. For severe PMDD/PMS, and in those with severe mood symptoms, drug therapy is warranted.
- SSRIs have a response rate 60-90% compared to 30-40% for placebo. Effective medications include Fluoxetine and Sertraline (these two are the most studied), Citalopram, Escitalopram, Clomipramine & Venlafaxine. Antidepressants reduce both mood and somatic symptoms; improve the quality of life and social functioning. Their effect on PMS is not just antidepressant effect as the effects appear within one menstrual cycle in most patients (thus too rapid for an antidepressant based mechanism). The most effective agent is considered to be fluoxetine.
- **Intermittent dosing** during the luteal phase of the menstrual cycle is also affective. This temporarily increases serotonin concentration during the luteal phase. Intermittent dosing is effective only in the luteal phase, 2 weeks prior to menses.
- A meta-analysis (Dimmock, 2000) reported an odds ratio of 6.91 (3.90 to 12.2) in favour of SSRIs compared with placebo. The placebo effect was large too. There was no significant difference in symptom reduction between continuous and intermittent dosing. But intermittent dosing is more effective than continuous or semi-continuous dosing (Wikander et al...1998), cheaper with less withdrawal related due to side effects. Disadvantages of intermittent dosing include lower efficacy for somatic symptoms compared to continuous treatment.

SSRI in Depression	SSRI in PMS
High frequency of sexual side effects	Low frequency of sexual side effects
Continuous dosing more effective	Intermittent dosing more effective
Effective in three -four weeks	Effective in few days
Receptor site difference than PMS	Receptor site difference than PMS
Akathisia and increased suicidal ideation reported	No reports of akathisia or increased suicidal ideation

- Alprazolam can be used with caution in premenstrual insomnia and overwhelming anxiety
- Hormonal treatment to suppress the ovulation-Long acting GnRH agonist, estrogen and certain new contraceptives; should be used only as a last resort because of the potential implications of introducing early menopause in these women
- Remission rates are low on cessation of treatment. PMS can last until the menopause.

B. Myocardial infarction and depression

- 20% of patients with coronary heart disease have comorbid depression, with a larger proportion experiencing subsyndromal depression (Ramamurthy et al., 2013).
- Psychological and behavioural interventions for patients with coronary artery disease reduce depression and mortality (Welton et al., 2009).
- Patients who have persistent depression, despite treatment, may be at increased cardiac risk (OR=1.53, Nicholson et al., 2006).
- Several clinical trials have been conducted to determine whether treating depression reduces the risk for cardiac events following a recent acute myocardial infarction: These include the Enhancing Recovery in Coronary Heart Disease (ENRICHD) study for CBT, the Myocardial Infarction and Depression Intervention Trial (MIND-IT), the Canadian Cardiac Randomized Evaluation of Antidepressant and Psychotherapy Efficacy [CREATE] for interpersonal therapy, a problem-solving therapy trial called the COPEs, and a trial evaluating CBT based stress management (Women's Hearts Study).
- In the primary analyses, both the ENRICHD and MIND-IT interventions had only modest effects on depression and neither of them improved survival.
- The Sertraline Antidepressant Heart Attack Randomized Trial (SADHART) is to date the largest randomized trial evaluating the use of antidepressant medication for depressed patients with heart disease. SADHART compared sertraline vs. placebo in a 16-week trial. No difference in safety was noted (change in left ventricular ejection fraction, an increase in premature ventricular contractions, or prolongation of the QT interval) between the treatment and placebo groups. A nonsignificant reduction in the composite endpoint (MI or CHD death) in the sertraline group (relative risk, 0.77; 95% confidence interval, 0.51-1.16) indicating that SSRIs may be directly cardioprotective by reducing platelet activation. But there was little difference in depression status between groups receiving sertraline and placebo after 24 weeks of treatment. However, the effect of sertraline was greater in the patients with severe and recurrent depression.
- **Heart failure and depression:** The prevalence rates of clinically significant depression among CHF patients is approximately 21.5%, (i.e., 2 to 3 times the rate of the general population). A higher prevalence is associated with higher NYHA functional class and in females. This high rate is similar to the rate of depression seen in CAD/Post-MI patients. It appears that the relative risk of mortality in patients with CHF who are depressed is about 2:1 when compared to the risk in non-depressed CHF patients. Rates clinical events, rehospitalization, and general health care use are markedly higher among CHF patients reporting more severe depression.

C. Specialist advice for endocrinology and metabolic disorders

Hyperthyroidism:

- Symptoms: sweating, fatigue, heat intolerance, weight loss, weakness, fine tremor, and tachycardia.
- Psychiatric symptoms – most commonly generalized anxiety; depression, irritability, hypomania, and cognitive dysfunction are also noted. In severe thyrotoxicosis, mania may occur.

Hypothyroidism:

- M: F ratio is 1:6 (source: Kumar & Clark Textbook of Medicine 6th edn p 1077)
- Symptoms: weakness, fatigue, cold intolerance, weight gain, constipation, and somnolence.
- Psychiatric symptoms - depression (often misdiagnosed as major depressive disorder and hypothyroidism may be missed), also cognitive dysfunction is noted. Psychosis may be seen in severe cases (myxedema madness).
- Subclinical hypothyroidism is a risk factor for depression and rapid cycling in bipolar disorder.

Hyperparathyroidism:

- Symptoms directly proportional to serum calcium levels. At mild-moderate (10–14 mg/dL) - depression, apathy, irritability, lack of initiative, and lack of spontaneity. If severe (>14 mg/dL) - delirious with psychosis, catatonia, or lethargy, progressing to coma.

Hypoparathyroidism:

- Symptoms directly proportional to serum calcium levels. In mild hypocalcemia - patients have anxiety, paresthesias, irritability, and emotional lability. If severe - mania, psychosis, tetany, and seizures may occur.

Cushing's syndrome: (Cortisol excess)

- Most commonly results from exogenous corticosteroids. ACTH secretion by a pituitary tumour is called Cushing's disease. Corticosteroid secretion by an adrenal adenoma is also a possible cause.
- Physical symptoms include diabetes, hypertension, muscle weakness, obesity, and osteopenia
- Psychiatric symptoms may appear before physical signs. Depression is most common, followed by anxiety, hypomania/mania, psychosis, and cognitive dysfunction. Exogenous steroid produces more mania than endogenous steroids. Psychiatric effects are often dose-related.

Addison's disease: (Cortisol depletion)

- Psychiatric symptoms - apathy, anhedonia, fatigue, and depressed mood.
- May be misdiagnosed as major depressive disorder. Anorexia is common in both, but the presence of nausea, vomiting and skin changes (dark pigmentation in some) should suggest Addison's disease.

Acromegaly:

- Occurs as a result of growth hormone excess.
- Psychiatric symptoms include mood lability, personality change, and depression. Psychosis may be due to treatment with dopamine agonists such as bromocriptine.

Pheochromocytoma:

- Results from a catecholamine-secreting tumour.

- Physically - tachycardia, labile hypertension, headache, sweating, and palpitations - episodic. Symptoms may mimic panic attacks.
- Screening by testing for urinary catecholamines (Vanillyl mandelic acid, metanephrines). The best diagnostic test is a plasma metanephrine level, which is more specific.

Diabetes:

- Depression is 2–3 times more common in diabetics than in the general population. Depressed diabetics have poorer glycemic control and increased diabetic complications
- An increased prevalence of diabetes type 2 is seen in patients with bipolar disorder, schizophrenia (2-4-times higher) and severe depression.
- Permanent cognitive dysfunction in diabetics rarely results from recurrent hypoglycaemia, but frequent hyperglycemic episodes results in cognitive dysfunction due to cerebral micro- and macrovascular damage.

D. Specialist advice for palliative care (from Dein, 2003)

- Prevalence of major depression in those with advanced cancer is 5–15%. Another 10–15% present with less severe symptoms. Somatic symptoms are not useful when diagnosing depression in such cases. Pervasive global anhedonia is a more useful criterion for the diagnosis of depression. SSRIs (not much evidence) low dose amitriptyline (avoid in those with higher risk of delirium; useful if neuropathic pain is present) lofepramine and rapid-acting psychostimulant such as dexamphetamine or methylphenidate (for patients who have only weeks of life) are useful.
- Delirium is common in palliative care settings. The prevalence is around 44% in cancer in-patients, rising to 62% shortly before death. It may be either hypoactive – sedated subtype or hyperactive agitated type. The aetiology is usually multi-factorial, and a specific aetiology is discovered in less than 50% of terminally ill patients with delirium. Though small dose of haloperidol or lorazepam is frequently used, they are not always successful in controlling the symptoms, especially in sedated subtype.

E. Specialist advice for renal medicine (from Phipps & Turkington, 2001)

Psychotropic prescription in renal disorder

Benzodiazepines should be used with caution. The half-life of diazepam remains unchanged in end-stage renal disease but its metabolite, desmethyldiazepam, may accumulate, causing excessive sedation. The half-life of lorazepam is increased from 8–25 hours in normal adults to 32–72 hours in end-stage renal disease. At a low level of renal function, lorazepam dosage should be reduced by 50% to avoid excessive sedation.

Antidepressants: Imipramine and amitriptyline can be given at their usual dosage, as renal impairment does not increase their half-lives. The half normal dose is used for citalopram in patients with renal impairment or in elderly. The half-life of paroxetine is considerably increased in severe renal impairment, requiring dosage reduction. The dosage of fluoxetine and fluvoxamine does not have to be reduced in the elderly or patients with renal impairment. Sertraline manufacturers do not recommend it in renal impairment

Antipsychotics: Haloperidol does not require dose reduction in renal impairment unless excessive sedation or hypotension occurs. Amisulpride is renally excreted almost exclusively. Hence, renal failure will be a relative contraindication to use this drug. Product monograph suggests alternate day dosing or dose reduction if no other alternatives are possible. Risperidone and its active metabolite 9-hydroxy-risperidone are substantially excreted in the urine so that in renal impairment the elimination half-life is prolonged.

Lithium is best avoided or given at low dosages.

- **Uraemic encephalopathy:** Occurs when the glomerular filtration rate (GFR) falls to 10% of normal. The rate of onset is proportional to the rate at which the GFR falls and. The effects cause problems in cognition, psychomotor activity and personality, with vomiting, restlessness, myoclonus and coma, leading to death. This condition is reversible by treatment of the underlying disease, dialysis or renal transplant
- **Dialysis disequilibrium syndrome:** It is a temporary clinical disorder that may occur during or after the first few dialysis treatments. It is more common among younger patients. The high incidence among those with pre-existing neurological problems such as cerebral trauma and a recent stroke. Mild symptoms include headache and restlessness, which may be followed by nausea, vomiting, hypertension, tremor, disorientation and seizures. The condition is now much less common with improved dialysis technology. Most of the symptoms produced are secondary to cerebral oedema.

E. Specialist advice for other medical disorders

- **Lyme's disease:** Lyme disease is caused by *Borrelia burgdorferi*, transmitted via wooden tick bite which normally lives on deers. A red spot develops initially around the location (crotch or armpit). This will develop a central clearing called *erythema migrans* within 4 weeks. About 15% develop neuroborreliosis, where CNS is affected. Back pain worse at night, facial numbness and facial palsy may develop. This is rare in the UK; more common in North America.
- Patients with late-stage Lyme disease may present with a variety of neurological and psychiatric problems, ranging from mild to severe. These include cognitive deficits especially memory impairment ("brain fog"), word-finding problems, visual/spatial processing impairment and slowed processing of information. Some patients may experience psychosis, seizures, violent behaviour and irritability along with anxiety and depression. Neuroborreliosis can mimic virtually any type of encephalopathy or psychiatric disorder and is often compared to neurosyphilis.
- **Psychiatric aspects of SLE (Systemic Lupus Erythematosus):** SLE, a prototype multisystem autoimmune disease, is characterized by non-organ specific vasculitis and the presence of anti-nuclear antibodies (ANA). It can be either acute or insidious in its onset, and has a chronic, remitting and relapsing, course of a febrile illness characterized by injury to the skin (butterfly rash in middle-aged woman), joints, kidney and serosa.
- Psychiatric symptoms include depression and anxiety, particularly in recently diagnosed patients and those with disfiguring skin lesions. Rarely, psychosis can also occur.
- CNS manifestations include peripheral neuropathy - including rarely, the Guillain-Barre Syndrome, grand mal seizures, chorea and choreoathetosis, cognitive impairment affecting memory, perception, orientation and intellectual function, severe headaches, stroke, B-cell lymphoma and limbic encephalitis-type picture.
- **Ca Pancreas - insulinoma:** Insulinoma is one of the most common neuro-endocrine tumors of the pancreas. Some cases with insulinoma present with neuropsychiatric symptoms and are often misdiagnosed as psychosis / depression. Patients with insulinomas usually have a recurrent headache, lethargy, diplopia, and blurred vision, particularly with exercise or fasting.

- **Neurosarcoidosis:** Idiopathic granulomas in various tissues – mainly lungs and mediastinal node. May affect CNS especially cranial nerves are producing bilateral facial palsy. Depression is seen in 20%. Rarely psychosis can occur. Patients may have erythema nodosum on their shin – a cutaneous sign. ACE levels in the blood are elevated due to macrophage activity. Treatment is with immunosuppression.
- **Metachromatic leucodystrophy:** “Metachromatic leukodystrophy (MLD) impairs the growth or development of the myelin sheath and is caused by genetic defects of the enzyme arylsulfatase A. MLD is one of several lipid storage diseases. There are three forms of MLD: late infantile, juvenile, and adult. In the late infantile form, which is the most common MLD, affected children have difficulty walking after the first year of life. Symptoms include muscle wasting and weakness, muscle rigidity, developmental delays, progressive loss of vision leading to blindness, convulsions, impaired swallowing, paralysis, and dementia. Children may become comatose. Most children with this form of MLD die by age 5. Children with the juvenile form of MLD (between 3-10 years of age) usually begin with impaired school performance, mental deterioration, and dementia and then develop symptoms similar to the infantile form but with slower progression. The adult form commonly begins after age 16 as a psychiatric disorder or progressive dementia. Adult-onset MLD progresses more slowly than the infantile form” . (Excerpt from NINDS website)
- Nearly 60% of adolescent onset MLD cases have schizophrenia-like psychosis – this is not seen in younger onset cases. Features of the adult form include mental deterioration, impaired concentration, depression and dementia.
- **Neuroacanthocytosis:** Genetically heterogenous neurologic disorders characterized with acanthocytosis. Neurologic problems usually consist of either movement disorders or ataxia, personality changes, cognitive deterioration, axonal neuropathy, and seizures. Acanthocytosis refers to a certain percentage of the patients' erythrocytes (typically 10-30%) having an unusual star-like appearance with spiky- or thorny-appearing projections. Personality changes include impulsivity, distractibility, anxiety, depression, apathy, loss of introspection, and compulsivity.
- A peculiar gait is characterized by lurching with long strides, and quick, involuntary knee flexion is seen. Seizures, generally tonic-clonic, are noted. Subcortical dementia may set in.

2. Psychiatric aspects of brain diseases

A. Multiple sclerosis:

- ★ Starts between the ages of 20 and 40. In the UK, the lifetime risk is 1:8000; It is twice as common in women as in men. Seen with greater frequency as the distance from the equator increases.
- ★ Characterized by multiple demyelinating lesions with a predilection for the optic nerves, cerebellum, brain stem and spinal cord. Diverse neurological signs reflect the presence and distribution of plaques. It is predominantly a white matter disease.
- ★ 5–10% of those affected show a steady progression of disability, with no remissions (primary progressive multiple sclerosis); 20–30% follow a relapsing–remitting course but never become seriously disabled; 60% enter a phase of progressive deterioration following a number of relapses and remissions (secondary progressive)
- ★ Steroids continue to play a key role in the treatment. Glatiramer acetate is used as a neuroprotective agent and an immunomodulator and is used to reduce the frequency of relapses in relapsing-remitting multiple sclerosis. It is sold under trade name Copaxone and is administered by subcutaneous injection at a dose of 20 mg per day. Cannabinoids are not licensed but may be available as a named patient basis.
- ★ **Depression:** A lifetime prevalence of depressive symptoms = 40–50%. This is about three times higher than the rate in the general population. Depression is linked with poorer cognitive functioning, poor compliance with MS treatment and a lower quality of life.
 - Drug-induced low mood is an important differential diagnosis. Steroids (more likely), baclofen, dantrolene and tizanidine can cause depression. Controversial association with *beta* interferon – note that it is Interferon *alpha* which is clearly associated with depression (used in viral hepatitis).
 - No clear association between brain abnormalities identified by MRI and depression
 - Desipramine & SSRIs have an evidence base for use in treatment; ECT may be used but 20% risk of triggering a relapse of multiple sclerosis if ECT is given. Presence of active brain lesions on MRI before treatment is a potential risk factor for MS relapse following ECT
 - 3% of people with multiple sclerosis die by suicide over a 6-year period, over 16 years, suicide accounted for 15% of all deaths. Suicidal ideation present cross-sectionally in nearly 30%.
- ★ **Mania:** Higher than expected prevalence of mania is seen. This can also be drug induced: steroids (more likely), baclofen, dantrolene, tizanidine (central muscle relaxant). Mild to moderate degrees of mania seen in up to 1/3rd of patients given steroids. People with multiple sclerosis who become hypomanic on steroid therapy are more likely to have a family or premorbid history of affective disorder and/or alcoholism.
- ★ **Psychosis:** There is MRI evidence suggesting that patients showing mania with psychotic symptoms have plaques that are distributed predominantly in the bilateral temporal horn areas
- ★ **Pathological laughing and crying syndrome:** Also called as emotional incontinence/emotionalism or pseudo bulbar affect. This refers to spontaneous, unprovoked laughter or crying seen in patients with any type of brain damage including MS; it occurs in response to non-specific stimuli, in the absence of associated mood change (mood-incongruent affect) and in the absence of voluntary control of facial

expression. The affect change is generally transient but distressing due to perceived loss of control.

- In a study, 66% of people with MS and pathological laughing / crying responded to 75 mg of amitriptyline per day. Improvement in symptoms also occurs with either amantadine or levodopa administration. Fluoxetine and sertraline have been evaluated in small parallel RCTs with favourable results; citalopram in a crossover trial showed significant benefits and fewer side-effect

- ★ **Cognitive impairment:** A significant extent of impairment in cognition is seen in some not all cases; may predate neurological lesions. A subcortical pattern of deficits has been noted; MMSE is not helpful for screening. Donepezil has been shown to improve memory in multiple sclerosis in a randomised clinical trial

B. Stroke:

Psychiatric disorders in stroke: (from Chemerinski & Robinson, 2000)

Syndrome	Prevalence
Post-stroke depression (more subcortical lesions)	35%
Post-stroke anxiety (more cortical lesions)	25%
Post-stroke mania	Rare
Apathy without depression	20%
Emotional incontinence (Pathological cry/laugh)	20%
Catastrophic reaction (burst of aggression / anxiety when faced with tasks)	20%
Psychosis	Rare

- ★ **Post-stroke depression:** The mean duration of post stroke major depression is 34 weeks. It was thought that left-hemisphere lesions are depressogenic; this has been contested recently. Major depressive disorders are common with infarcts involving the basal ganglia, especially on the left hemisphere. National clinical guidelines for the management of mood disturbance after stroke (Intercollegiate Working group, 4th edn; 2012):
 - Screen for depression and anxiety within the first month of stroke; Confirm emotionalism by a few simple questions at interview; If one mood disorder is present assess for the others
 - In mild to moderate depression, support to increase social interaction, exercise and other psychosocial interventions are recommended.
 - Severe, persistent or troublesome tearfulness (emotionalism) should be treated with anti-depressants; the frequency of crying should be monitored to check effectiveness
 - Consider a trial of antidepressant medication in those with persistently depressed mood; if there is a good response, antidepressants should be continued for at least 4 months after initial recovery. Both fluoxetine and citalopram have RCT evidence to support their use though fluoxetine has not

been consistently effective in trials. Brief, structured psychological therapy should be considered for patients with depression.

- Antidepressant treatment should not be used routinely to prevent the onset of depression

C. Epilepsy:

Symptoms	Frequency
Depression	30–50%
Panic disorder	20%.
Psychosis	3 to 7%
(from Dilley & Fleminger, 2006)	

- ★ **Depression in epilepsy** is associated with stigma and demoralization. It is also linked to with location of lesion – esp. common in TLE. High risk of suicide is present in epilepsy even without depression (10-15% in certain cohorts); mortality shoots when clinical depression is positive (nearly 25 times higher). SSRIs are the first-line; may reduce seizure threshold but rare. Lithium can increase seizure severity and reduces the seizure threshold.
- ★ **Panic: attacks** can be interictal or peri-ictal. These must be differentiated from seizure activity
- ★ **Psychosis is more** common in partial epilepsies. Possible role of mesial temporal and extratemporal damage as a risk factor. Inconclusive evidence for increased risk of psychosis in temporal lobe epilepsy or with left-sided focus. Episodic psychosis is most commonly post ictal – may have affective and confusional components – visual hallucinations are commoner than in functional psychosis. Chronic interictal/schizophrenia-like psychoses of epilepsy may be a separate disorder.
 - Psychosis may also be drug related and lasts for less than a week; most antiepileptics have psychiatric side effects; vigabatrin may be especially associated with psychosis. Sulpride and haloperidol may be less epileptogenic than other antipsychotics
- ★ **Pseudoseizures (Non-epileptic seizures)** are associated with past psychiatric history, somatisation, and social stressors – esp. abuse in childhood. Pseudoseizures more likely to happen in the daytime and when others are present. Patients are less likely to sustain injury from seizures; side-to-side head movements may be seen; seizures may be more prolonged than usual. Eyes may be kept tightly shut during a pseudoseizure while true seizures can occur with open eyes. More likely to maintain body tone during a pseudo-seizure. Regaining of alertness and orientation is rather rapid. Ability to recall events clearly is preserved in pseudoseizures; crying or emotional displays may be often seen in pseudoseizures. *Incontinence* has no diagnostic value. An *abnormal EEG* does not always mean epilepsy. Increased post-ictal serum prolactin concentrations (>1000 IU/l) despite normal baseline levels are found after epileptic seizures but measurements must be taken within 15 min of the event.

D. Parkinson's disease:

Psychiatric manifestations	Frequency
All psychiatric symptoms	70%
Depression	40-50%
Hypomania/euphoria	2%/10%
Anxiety	50-65%
Apathy	Common
Psychosis	40% (drug-related)
Cognitive impairment	19% with no dementia; 25-40% dementia
(from Dillely & Fleminger, 2006)	

- ★ **Depression:** Risk factors include female gender, younger onset, presence of prominent right-sided lesions, bradykinesia and gait disturbance, rapid disease progression, poorer cognitive status and activities of daily living
- ★ **Apathy without depression** is also common and is associated with executive dysfunction. May be masked by Parkinsonian affect – subjective exploration may be necessary.
- ★ **Mania:** Denovo mania is rare in Parkinson's. Mania is mostly associated with levodopa and dopamine agonist treatment especially in pre-existing bipolar affective disorder/ family history of bipolar disorder. Levodopa can also result in pathological gambling, hypersexuality and hallucinations in addition to hypomania.
- ★ **Psychosis** is mostly drug related (both dopaminergic/anticholinergic drugs). Hallucinations are common 20% – especially visual modality; delusions seen in 3–30%. Low-dose (<100 mg/day) clozapine can improve psychosis without worsening parkinsonism. Quetiapine can also be used in the treatment.
- ★ **Cognitive impairment** is common; poor prognosis if present at time of initial referral. Ranges from mild to severe dementia syndrome. Risk factors include older age, late-onset Parkinson's disease, low socio-economic status and education and the presence of severe extrapyramidal signs. The incidence of hallucination and delusions appear to be higher than in Alzheimer's. Frontal executive dysfunction with a subcortical pattern of dementia is seen in most patients. Neuroleptic sensitivity is also seen. Lewy body dementia is pathologically related closely to Parkinson's dementia; rivastigmine is useful in LBD. No drugs licensed yet for Parkinson related dementia.

E. Huntington's disease:

- ★ Progressive neurodegenerative disorder with chorea and dystonia, incoordination, cognitive decline, and behavioural difficulties with autosomal dominant high penetrance pattern of inheritance. The onset of symptoms is often in middle-age after affected individuals have had children. The disorder can manifest at any time between infancy and senescence. Prevalence of psychiatric symptoms at the first presentation = 30%. Suicide rates 4 times higher than the general population (note unaffected family members also have higher suicide risk, sometimes as high as patients). 3–6% of cases a **schizophreniform psychosis** is the first presentation of Huntington's disease. OCD like symptoms

may occur due to basal ganglia involvement. MRI shows caudate atrophy (head) and resultant box-car ventricles (from Dilley & Fleming, 2006; Walker, 2007).

- ★ The mutant protein in Huntington's disease—huntingtin—results from an expanded CAG repeat leading to a polyglutamine strand of variable length at the N-terminus. This tail confers a toxic gain of function. The gene for Huntington's disease (HD) is located on the short arm of chromosome four and is associated with an expanded trinucleotide repeat. Normal alleles at this site contain CAG repeats, but when these repeats reach 41 or more, the disease is fully penetrant. Incomplete penetrance happens with 36–40 repeats, and 35 or less are not associated with the disorder. Inheritance shows the phenomenon of anticipation. The number of CAG repeats accounts for about 60% of the variation in age of onset. Though low penetrance 36-39 CAG repeats are identified, these are relatively rare, and penetrance is calculated to be around 95%.

F. Wilson's disease:

- ★ An exclusively psychiatric presentation occurs in 20% of cases. 50% of patients will have mental disturbances at some point during the course of the disease. Personality disturbances, mood abnormalities, and cognitive dysfunction being the most common psychiatric symptoms. Psychiatric manifestations tend to occur with neurological forms of WD rather than with hepatic ones. Cognitive impairment occurs in up to 25% of patients. A frontosubcortical pattern of dementia is noted. Depression occurs in 30% of cases of WD. Suicidal behaviour may occur in between 4% and 16%. Mania can occur but is less frequent than depression. Psychosis has been described in WD and can indeed be the initial presentation, but its frequency is very low at about 2% with WD (from Ring & Serra-Mestres, 2002).
- ★ **Kayser-Fleischer rings** (copper deposits on Descemet's membrane of the cornea) are present in 95% of patients with neurological symptoms, in 50-60% of patients without neurological symptoms and in only 10% of asymptomatic siblings.
- ★ MRI may show intense hyperintensity of the midbrain with relative sparing of the red nucleus, superior colliculus and part of the pars reticulata of the substantia nigra with hypointensity of the aqueduct sometimes called as "Giant Panda sign".
- ★ Specific treatment of WD is with copper chelating or copper depleting agents. Most neurological and psychiatric manifestations of WD can improve with this treatment; however, the early diagnosis and initiation of the treatment is essential.

G. Transient Global Amnesia (TGA):

TRANSIENT GLOBAL AMNESIA: Diagnostic criteria
Attacks must be witnessed, and information must be available from a capable observer who was present for most of the attack
There must be clear-cut anterograde amnesia during the attack
Clouding of consciousness and loss of personal identity must be absent, and the cognitive impairment must be limited to amnesia (e.g., no aphasia, apraxia)
There should be no accompanying focal neurologic symptoms during the attack and no significant neurologic signs afterwards
Epileptic features must be absent
Attacks must resolve within 24 hr
Patients with recent head injury or active epilepsy (i.e., currently on medication or at least 1 seizure in the past 2 yr) are excluded

Transient global amnesia is rare and is seen at a rate 5 to 10/100,000 per year; the rate rises to 30/100,000 in those older than age 50. Though the aetiology is unclear, most cases involve hypoperfusion in the temporal and parietotemporal regions, particularly in the left hemisphere. TGA is characterized by the abrupt onset anterograde amnesia characterised by significant new learning deficit. Patients show mild confusion and a lack of insight into the problem but an intact sensorium. Episodes last from 6 to 24 hours. Most patients show complete improvement though recurrence is high.

H. Fahr's disease:

- ★ Idiopathic progressive calcium deposition in the basal ganglia with onset between the ages of 20 and 40 being associated with schizophreniform psychoses and catatonic symptoms, and onset between the ages of 40 and 60 being associated with dementia and choreoathetosis.
- ★ 50% had psychiatric problems (from Ring & Serra-Mestres, 2002). Psychiatric symptoms correlate with more extensive calcification. Depression is also very common, but mania much less so. The pattern of cognitive impairment found in FD is of the frontosubcortical type. The commonest neurological features of FD are Parkinsonism, chorea, dystonia, tremor, gait disturbance, dysarthria, seizures, and myoclonus.
- ★ Fahr's syndrome is the term used to describe specific causes of calcium deposition in the basal ganglia such as hypoparathyroidism.
- ★ Radiological basal ganglia calcification without clinical features on CT may occur at a rate of about 0.9% MRI shows hypointensity of the striatum. EEG is non-specific – may show diffuse changes.

I. Herpes simplex encephalitis:

- ★ The commonest identified cause of viral encephalitis and the commonest cause of limbic encephalitis – affects temporal lobe and limbic circuit. 70% of cases are caused by herpes simplex type 1. Immuno-compromised patients may have herpes simplex type 2, human herpes viruses (HHV) 6 or 7.

- ★ A fairly abrupt onset of confusion, memory impairment, and often seizures. 70% show psychiatric disturbances including acute confusion, depression and psychosis. Fever is common but not invariable.
- ★ Neuroimaging usually reveals signal change and swelling within the temporal lobes; this is often visible on CT but is more easily seen using MRI. Significant brain swelling may lead to raised intracranial pressure. Cerebrospinal fluid (CSF) examination commonly reveals a lymphocytosis and raised protein. The gold standard for in vivo diagnosis is CSF polymerase chain reaction (PCR) for herpes viruses, which has a sensitivity and specificity of ~95%.
- ★ Magnetic resonance imaging (MRI) provides the most sensitive method of detecting early lesions and is the imaging of choice in HSE. The EEG is invariably abnormal in HSE. Non-specific slowing may be the only feature early in the illness; the later stages are more likely to be associated with high voltage periodic lateralizing epileptiform discharges. If present they are diagnostically useful but they are not specific for HSE
- ★ Untreated, there is a case fatality of ~70%. Treatment with intravenous aciclovir reduces case fatality to 20–30%. Treatment in proven cases should continue for at least 14 days, longer in immuno-compromised patients (From Kennedy & Chaudhri, 2002; Schott, 2006).
- ★ HSE is the most common cause of *Kluver-Bucy syndrome* in humans; this is characterized by emotional blunting, hyperphagia, visual agnosia and inappropriate sexual behaviour. This is due to bilateral temporal lobe damage – and most often only a partial syndrome is evident. Carbamazepine may be used to control some symptoms.

J. Meige Syndrome:

Henri Meige described in 1904 a new type of oral facial dystonia characterised by repetitive blinking and chin thrusting. Some patients have lip pursing or tongue movements and, for a few, the movements spread into the shoulders. Primary Meige's is idiopathic; antipsychotics, levodopa and Lewy body dementia can cause secondary Meige's. At times, there is a joint interactive movement between the oral movements and the eye movements. The patients are more likely to be women than men and usually at middle age. There is some variation with stress, but the movement is present both at rest and when active; it disappears with sleep. Patients may chew gum, whistle or touch their face in an effort to lessen the movements.

K. Traumatic brain injury:

- ★ Peak incidence of head injury is between the ages of 15 - 24 years
- ★ **Concussion** damage causes transient coma for hours followed by apparent complete clinical recovery. Brain **contusion** leads to prolonged coma, focal signs and lasting brain damage. Pathological support for this division is poor.
- ★ Mechanisms of TBI: Axonal and neuronal damage from (1) shearing and rotational stresses on decelerating brain, often at sites opposite impact (contrecoup effect) (2) damage from direct trauma (3) brain oedema and raised intracranial pressure (4) brain hypoxia and ischaemia.
- ★ Two types of amnesia can occur after head injury:

- **Post-traumatic amnesia (PTA)** includes amnesia for the period of injury and the period following injury until normal memory resumes. This is anterograde.
- **Retrograde amnesia** includes dense amnesia for the period between the last clearly recalled memory prior to the injury and the injury itself. Generally in minutes, it reduces with time gradually.
- ★ Poor **prognostic factors** with respect to psychiatric morbidity following head injury includes long duration of loss of consciousness, long PTA, elderly, chronic alcohol use, diffuse brain damage, new onset seizures and focal damage to dominant lobe.
- ★ GCS at 24 hours after injury is widely used to assess severity.
- ★ Apart from GCS (Glasgow coma scale) alternative indices of TBI severity such as **length of coma** (LOC), duration of **post-traumatic amnesia** (PTA), and the **Abbreviated Injury Scale** (AIS) have also been used to determine injury severity and establish prognosis for recovery
- ★ LOC and PTA have been used exclusively to predict **functional outcome**, but the AIS has been used to predict survival
- ★ Most investigations have found LOC or PTA to be more predictive of functional status than GCS.

Duration of PTA	Classification	Functional outcome
PTA less than 60 minutes	Mild injury	May return to work in <1 month
PTA between 1-24 hours	Moderate injury	May return to work in 2 months
PTA between 1-7 days	Severe injury	May return to work in 4months
PTA greater than 7 days	Very severe injury	May require > 1 year for return to work

- ★ **Late sequelae:**
 - Cognitive impairment is common especially after closed head injuries with PTA lasting >24 hours.
 - Personality changes are most likely after a head injury to the orbitofrontal lobe or anterior temporal lobe.
 - Depression (most common sequelae) and anxiety occur in roughly 1/4 of head injury survivors. A meta-analysis showed that the proximity of the lesion to left frontal lobe predicted depression. Suicide risk is also higher post head injury.
 - A schizophrenia-like psychosis with prominent paranoia is associated with left temporal injury while affective psychoses (esp. mania in 9% patients) are associated with right temporal or orbitofrontal injury.
 - There is also an increased prevalence of schizophrenia post head injury (-2.5% develop the disorder).
 - Post-traumatic epilepsy is seen in 5% closed and 30% open head injuries (usually during the first year) and worsens prognosis.
 - Less psychopathology in children after head injury due to increased brain plasticity.
- ★ **Post-concussion syndrome: (from King, 2003)**
 - Very controversial – some deny its existence.

- It follows mild head injury (post-traumatic amnesia <1h, Glasgow Coma Scale score 13–15, loss of consciousness (LOC) <15 min) or moderate head injury (post-traumatic amnesia 1–24h, Glasgow Coma Scale score 9–12, LOC 15 min to 6 h)
- Symptoms: headache, dizziness, fatigue, poor memory, poor concentration, irritability, depression, sleep disturbance, frustration, restlessness, sensitivity to noise, blurred vision, double vision, photophobia, nausea
- and tinnitus (King, 1997).
- At least 50% experience some post-concussion symptoms that recover completely within 3 months of injury, except in nearly a third.
- Evidence of organic aetiology includes post-mortem evidence of diffuse microscopic axonal injury after mild head injury. Macroscopic brain lesions seen in 8–10% of individuals who have had routine neuroimaging in the first week. Predominant lesions noted in frontal, temporal and deep white-matter areas. These do not correlate well with symptoms; usually resolve over 3 months. In addition, slower recovery is seen with increased age > 40 years. Poorer outcome seen if the past history of head injury, alcohol or substance misuse is present. A consistent correlation between the severity of symptoms and the severity of neuropsychological impairments in speed of information processing is noted.
- Evidence supporting for non-organic aetiology includes the observation of worse outcome where there is a pre-existing psychological disorder. The best early predictors of persisting post-concussion symptoms are psychological factors. High rates of comorbidity of anxiety and depression symptoms with post-concussion symptoms. Stress exacerbates post-concussion syndrome ; further there is a higher prevalence of postconcussion syndrome in women. There is an association between severity of post-concussion symptoms or time off work after mild head injury and seeking compensation; twice as many patients seeking compensation have post-concussion symptoms compared with those who are not. But few show significant improvement following settlement, even a year afterwards.
- There is good evidence from RCTs that early intervention within the first few weeks of mild head injury does significantly reduce post-concussion symptoms and limit the emergence of persisting problems. A single hour-long assessment and treatment session is usually sufficient; education plus reassurance has a superior outcome to education alone
- There are virtually no systematic follow-ups of mild head injury and post-concussion beyond 1 year after injury.

3. Pain, Fatigue and Sleep

A. Sleep disorders

International Classification of Sleep Disorders ICSD

Dyssomnias are primary sleep disorders, which cause either difficulty getting off to sleep, or remaining asleep (insomnia) or excessive sleepiness during the day. They are divided into
<i>Primary insomnia</i> <i>Primary hypersomnia</i> <i>Circadian sleep disorders</i> <i>Narcolepsy</i> <i>Breathing related sleep disorders</i> <i>Sleep state misperception</i>
Parasomnias (disorders which intrude into the sleep process) are subdivided according to the phase of sleep with which they are associated:
Arousal disorders (arising from NREM sleep) <i>Confusional arousals</i> <i>Sleepwalking</i> <i>Sleep terrors</i>
Sleep– wake transition disorders <i>Sleep starts</i> <i>Sleep talking, etc</i>
REM sleep parasomnias <i>REM behavioural disorder</i> <i>Nightmares</i> <i>Sleep paralysis</i>
There are also other parasomnias are those which do not fall into these three categories such as <i>Sleep bruxism</i> <i>Sleep enuresis</i>
ICSD Sleep-related movement disorders <i>Restless leg syndrome (RLS)</i> <i>Periodic limb movement disorder (PLMD)</i> <i>Sleep-related bruxism</i>

ICD-10 Sleep disorders:

Nonorganic insomnia
Nonorganic hypersomnia
Nonorganic disorder of the sleep-wake schedule
Sleepwalking [somnambulism]
Sleep terrors [night terrors]
Nightmares

Circadian sleep disorders: Alterations of the sleep-wake cycle may be a sign of circadian rhythm disturbances, such as those caused by jet lag and shift work. In delayed sleep phase syndrome (most common circadian problem), the patient is unable to fall asleep until very early morning. As time progresses, the onset of sleep becomes progressively delayed.

Narcolepsy (from Dilley & Fleminger, 2006): A prevalence of 0.025% is seen in the general population. Symptoms include excessive daytime sleepiness, sudden sleep attacks (narcolepsy) – the sleep is refreshing as it is REM in nature, cataplexy (seen in 75% of patients), sleep paralysis (seen in 30%), hypnagogic hallucinations (NOTE: only 10% of individuals present with all four phenomena.). Automatic behaviours reported in around one-third (complex behaviours that are not recalled when they regain alertness). The overall sleep duration per 24 hours is NOT increased in narcolepsy due to reduced and fragmented nocturnal sleep.

It can be either familial or sporadic but a strong association with an HLA-DQB1*0602 marker in almost all individuals regardless of ethnicity. Abnormally low concentrations of hypocretin-1 (orexin-A) is noted in CSF. Sleep polysomnogram often shows sleep latency of less than 10 min along with sleep-onset rapid eye movement (REM) periods. A multiple sleep latency test is also helpful in supporting the diagnosis. Stimulants such as methylphenidate or the newer modafinil can be used. Imipramine may have effect on cataplexy.

Obstructive sleep apnoea: Obstructive sleep apnoea is more common in men (4%) than in women (2.5%). History of loud snoring, breathing pauses, mouth breathing, restless sleep, and increased perspiration at night is often noted. Snoring is the most common presenting symptom. Other symptoms include excessive daytime sleepiness, morning headaches, and behavioural changes (paradoxical hyperactivity or depression, anxiety, emotional lability). Right-sided cardiac failure may occur in the untreated.

Sleepwalking: Sleepwalking is described as partial arousal from sleep during slow-wave stages 3 and 4. It is most common during the initial third stage of the sleep period.

Night-terrors: Night terrors are recurrent episodes of abrupt awakening from sleep characterized by a panicky scream, with intense fear and autonomic arousal. The individual usually has no recall of the details of the event and is unresponsive during the episode. Night terrors occur during the first third of the night, during stages 3 and 4 of NREM sleep.

REM sleep behavioural disorder: Normally REM sleep is associated with loss of muscle tone and dreaming. In RBD, there is no loss of muscle tone, and the dreams are acted as complex behaviours. Patients act out their dreams, with limited awareness of surroundings and may have violent or dangerous consequences that result in physical harm. The episodes arise during the middle to the latter third of the night during REM sleep. It is associated with loss of the normal atonia of REM sleep. It may occur idiopathically or associated with disorders such as Parkinson's disease, diffuse Lewy body disease, multiple system atrophy and Gullian-Barré syndrome. RBD may be the prodrome of neurodegenerative

diseases, such as DLB or Parkinson disease. It can precede the diagnosis of a movement disorder by several years. It is likely that the associated lesions are situated in the brainstem.

Treatment with clonazepam has been shown to be effective; making the sleeping environment safe is first-line measure. According to the ICSID the diagnostic criteria include movements of the body or limbs associated with dreaming and at least one of the following criteria: potentially harmful sleep behaviour, dreams that appear to be acted out, and sleep behaviour that disrupts sleep continuity (**Dilley & Fleminger, 2006**).

Restless legs syndrome: RLS presents with the following four diagnostic criteria in patients aged 12 or older: Akathisia (distressing, irresistible need to move legs), usually accompanied by paresthesias. (This is a core feature.); Motor restlessness; Symptoms worsen at rest; Symptoms worsen at night.

- RLS is often an awake phenomenon. It is a neurological sensorimotor disorder characterized by unpleasant sensations in the legs that preclude a smooth transition from wakefulness to sleep. These symptoms can also significantly interfere with personal or social evening activities. Patients will have excessive daytime sleepiness. Up to 80% to 90% of RLS patients can present with periodic limb movements during sleep (PLMS).
- **The prevalence of RLS** to be from 3% to 15% of the general population (median 7.2%). M: F = 1:2 (twice as prevalent in women as in men). A familial pattern is seen in more than 50% of patients with primary RLS.
- **Predisposing factors include** iron deficiency conditions (anemia, renal insufficiency, pregnancy). This is best evaluated by ferritin or iron saturation testing. Peripheral neuropathy, use of sedating antihistamines, centrally active dopamine-receptor antagonists (metoclopramide, prochlorperazine), antipsychotics, caffeine and most antidepressants (except bupropion) can also increase the risk.
- **Treatment includes** improvement of sleep hygiene, relaxation techniques, dopaminergic agents (Nonergot D2 agonists, e.g., ropinirole, pramipexole are more effective and better tolerated than the ergot derivatives such as pergolide, Bromocriptine and the dopaminergic precursors such as levodopa/carbidopa), anticonvulsants (gabapentin, CBZ), opioids (oxycodone, propoxyphene), and clonazepam. The drug first licensed was ropinirole; however, several other medications have been used “off-label” for years.
- Therapy should be geared toward control of symptoms of restlessness rather than symptoms of insomnia or excessive daytime sleepiness (the latter two symptoms should improve with specific RLS treatment). Oral iron supplementation should be prescribed if ferritin serum level is low. Opioids for patients with pain and severe symptoms, and benzodiazepines (diazepam, clonazepam) have been used with moderate success.

Periodic Limb Movement Disorder: PLMD is characterized by periodic episodes of repetitive and stereotyped limb movements that occur during sleep. These movements can cause clinical sleep disturbance expressed by insomnia or excessive daytime sleepiness. In contrast to RLS, PLMD is not an awake phenomenon but an *asleep* phenomenon, and its diagnosis requires polysomnographic

documentation of an increased number of PLMS in association with significant disruption of sleep architecture and symptomatology.

- Polysomnographic diagnostic criteria for PLMS1 are based on electromyography (EMG) of the right and left anterior tibialis muscles. The movements may or may not be followed by arousal or awakening as determined by the polysomnogram.
- PLMS are a common phenomenon in otherwise healthy individuals. They also become more prevalent with increasing age (occurring in up to 34% of people over 60 years of age). PLMS symptoms are also very common in other sleep disorders, including RLS (80-90%), narcolepsy (45-65%), and REM sleep behaviour disorder (70%).
- Dopaminergic impairment and iron deficiency have been implicated in the generation of PLMS, similar to RLS.
- PLMD patients should be given general recommendations for improvement of sleep hygiene. Pharmacologic treatment for PLMS is not warranted unless there is obvious sleep disruption. Treatment recommendations are similar to RLS regimens.

Bruxism (persistent grinding of the teeth): Bruxism is considered as a stereotyped movement disorder or rhythmic disorder. It is more frequent during the early part of sleep and may be related to stress and/or anxiety or dentition abnormalities or stimulants use. Bruxism is not limited to sleep but may also occur while the child is awake. Basal ganglia dysfunction has been hypothesized.

Sleep disturbances in psychiatric disorders

Initial Insomnia

- Characterized by difficulty falling asleep, with increased sleep latency (time between going to bed and falling asleep). Initial insomnia is frequently related to anxiety disorders.

Middle Insomnia

- Refers to difficulty maintaining sleep. Decreased sleep efficiency is present, with fragmented unrestful sleep and frequent waking during the night. Middle insomnia may be associated with medical illness, pain syndromes, or depression.

Terminal Insomnia

- Referred to as early morning waking, patients consistently wake up earlier than needed. This symptom is frequently associated with major depression.

B. Chronic Fatigue Syndrome

CFS is known by different names such as myalgic encephalomyelitis” “immune dysfunctional syndrome,” “post viral fatigue syndrome” “chronic mononucleosis syndrome.

Epidemiology: Prevalence: nearly 0.5%: US community sample – 0.23 % - 0.42 %. UK sample - 2.6%, reduced to 0.5% after exclusion of co morbid psychological diseases. Seems to be higher in women M-F – 1:3, those belonging to lower occupational status and lesser educational background. The mean age of onset is between 29-35 years. Mean illness duration – 3-9 years. Prevalence is lower among children and adolescents – but still it is diagnosed in these age groups. The epidemiological studies are difficult to compare due to the lack of universal definition and different study methods. The quality of these epidemiological studies is poor.

CFS criteria US-CDCP (1994): Characterized by persistent or relapsing unexplained chronic fatigue of new onset, lasting at least 6 months and not the result of an organic disease or of continuing exertion and not alleviated by rest. Four or more of the following symptoms, concurrently present for more than 6 months: impaired memory or concentration, sore throat, tender cervical or axillary lymph nodes, muscle pain, pain in several joints, new headaches, unrefreshing sleep, or malaise after exertion.

Exclusion criteria include major psychiatric disorders such as psychotic depression, bipolar disorder, Schizophrenia, dementia, eating disorders, alcohol or substance abuse. Note: Non psychotic psychiatric disorders are not exclusion criteria for the diagnosis of CFS

The core symptom of CFS is fatigue. However, pain and cognitive difficulties can be as prominent as fatigue. Almost all patients develop functional impairment. Social relationships are affected in all the patients. 33% are unable to work, and 33% can work only part time. According to NICE, symptoms that might indicate another serious illness include significant weight loss; clinically significant lymphadenopathy; localising or focal neurological signs; features of inflammatory arthritis, connective tissue disease, or cardiorespiratory disease; and sleep apnoea.

Onset is sudden and noticed after an episode of viral infection especially in patients presenting to tertiary centres. In the general population in most patients with CFS, fatigue is gradual in onset. Hence, they may

PREDICTORS OF POOR OUTCOME

(Powell et al, 2004)

1. *Claiming a disability related benefit*
2. *Low sense of control*
3. *Strong focus on physical symptoms*
4. *Being passive with reduced activity*
5. *Membership of self help group (does not mean self help groups worsen CFS, membership may be an indicator of perceived severity)*

Factors not useful as predictors of treatment response:

1. *Severity of symptoms*
2. *Duration of illness*

not present for treatment. Patient with tender lymph nodes and sore throat are over-represented in clinical studies as they tend to present to the medical team more frequently.

Aetiology: The aetiology is probably multi factorial and is best understood by dividing it into predisposing (neuroticism, childhood inactivity or illness), precipitating (Infectious mononucleosis, Q fever, Lyme disease or serious life events in some) and perpetuating factors (strong belief in physical cause, activity-avoidance, poor self-control, primary/secondary gains and low self-perception of cognitive ability). CFS patients have an abnormality in HPA-axis and serotonin pathway suggesting altered physiological response to stress. This is the most consistent finding in CFS studies. 1/3rd have hypocortisolism. This is not a primary adrenal insufficiency; rather abnormality originates from CNS. Family studies have suggested mutation of cortisol transporting globulin.

CBT and graded exercise therapy are the only effective evidence-based treatments for CFS.

CBT for CFS: CBT is not a cure for CFS. However, its effectiveness can last at least up to five years in those who respond (Deale et al., 2001). CBT targets the cognitive and behavioural elements of CFS. The current evidence suggests that misperception of the cause of illness, avoidance of activity, excessive rest perpetuate the symptoms of CFS. The timing of intervention seems to be very important factor in the usefulness of CBT. Patients improved when medications were added to CBT and not when CBT was added to medications in that order (Stubhaug et al., 2008).. This study highlights the importance of the timing.

Graded exercise therapy: It is based physiological model of deconditioning. Muscles strength, autonomic response and perception of exercise related sensations are affected by inactivity. Therapy aims to gradually increase the exercise and other activities thus reducing the unwanted consequences of inactivity. Differences between CBT and GET

CBT	GET
Based on behavioural mode of avoidance	Based on physiological model of deconditioning
Always include a graded activity programme	Cognition not specifically treated. Some studies do cognitive work to encourage graded exercise
70% improvement rate	55% improvement rate
More complex training needed	Less complex training

Antidepressants should not be used routinely in CFS. If needed, it should be guided by the severity of depressive symptoms Following factors should be considered when using antidepressant – prolonged

COMPONENTS OF CBT for CFS (Prins et al, 2006)

Explanation of aetiological model

Motivation for CBT

Challenging and changing of fatigue related cognition

Achievement and maintenance of basic amount of physical activity

Gradual increase in physical activity

Rehabilitation e.g., rigorous self-monitoring, a safety behaviour in social phobia, can feed to the core symptoms.

inactivity increases the risk of autonomic side effects like postural hypotension and sedation may worsen fatigue.

Pacing: It is a strategy in which people with CFS are encouraged to achieve a balance between rest and activity. This involves having some limited types of activity alternating with periods of rest. The aim is to prevent a 'vicious circle' of overactivity and setbacks ('boom and bust') while setting realistic goals for increasing activity when appropriate. Theoretically it is based on conserving energy to be used wisely. Pacing is recommended by patient groups. But its effectiveness is questionable. It might prolong illness by encouraging avoidance. NICE includes pacing based on consensus recommendations as one of many alternatives that could be tried.

Prognosis: Duration of most prognostic studies range from 1-5 years. Full recovery without treatment is rare. 17-65% patients improve over 5 years. Less than 10% recover over 5 years. 10-20% worsens at the same time. CFS is not associated with increased mortality

CFS & Depression: There is a high rate of depression in patients with CFS. 23% of CFS patient have current major depression and 50%- 75% have life time history of major depression (Wessley et al., 1998).

	CFS	Depression
Serotonergic symptom	CNS upregulation	CNS downregulation
Symptoms	Absence of lack of motivation, guilt, anhedonia	Absence of sore throat, adenopathy, post-exertional fatigue
HPA axis	Central downregulation	Central upregulation
sleep	Typical sleep disturbance of depression	Reduce REM latency and increased REM density
antidepressant	Not effective	effective

CFS and anxiety disorder: Life prevalence of panic disorder in CFS 17-25%. Life time prevalence of generalized anxiety disorder in CFS 2-30%. There are overlapping symptoms between anxiety disorder and CFS. Both conditions have decreased cerebral blood flow sympathetic overactivity and sleep abnormality. However, CFS is not a physical presentation of anxiety disorder.

CFS and somatisation: Rate of somatisation in CFS is 28%. The diagnosis of CFS in somatisation is based on examiners attribution of CFS symptoms. If examiner attributes symptoms to the physical cause, the rate of somatisation decreases; if they attribute it to psychological cause then the rate of somatisation increases.

Other overlapping conditions: 20-70% of patient with fibromyalgia meet criteria for CFS, and 35-70% of those with CFS meet criteria for fibromyalgia. Symptoms of CFS can overlap with fibromyalgia, multiple chemical sensitivity, irritable bowel syndrome, temporomandibular joint disorder.

C. Psychiatric aspects of chronic pain

Pain as a major psychiatric disorder: The diagnostic criteria for pain disorder in DSM-IV-TR has been replaced by a new diagnostic criteria called Somatic Symptom and Related Disorders (SSD) in the DSM-5. This description insists that the diagnosis is made “on the basis of positive symptoms and signs (distressing somatic symptoms plus abnormal thoughts, feelings, and behaviours in response to these symptoms) rather than the absence of a medical explanation for somatic complaints.” “The DSM-IV disorders of somatization disorder, hypochondriasis, pain disorder, and undifferentiated somatoform disorder have been removed, and many, but not all, of the individuals diagnosed with one of these disorders, could now be diagnosed with SSD.

Pain associated with other psychiatric disorders: Psychiatric illnesses may increase pain intensity via a shared-mechanism through central pain modulation system. Treatments that provide substantial improvements in depression lead to moderate reductions in pain severity and disability. Pain is the most common presenting somatic symptom in medical outpatients; depression is the most common mental disorder in found in 10% to 15% of all those who suffer from pain disorders; substance misuse and post-traumatic stress disorder are also commonly seen (Geisser *et al.*, 1996). 43% of depressed adult patients report pain (Ohayon & Schatzberg 2003; based on telephone survey of a large European sample)

CBT appears to be effective for chronic pain when the following components are included - cognitive restructuring; relaxation training; time-based activity pacing; and graded homework assignments to decrease avoidance and to encourage a more active lifestyle.

Atypical Facial pain: The term atypical facial pain was first introduced by Frazier and Russell in 1924. The pain is atypical in its distribution, unilateral and poorly localised – not fitting with any cranial neuralgia. It lasts most of the day and is described as a severe ache, crushing or burning sensation. The Headache Classification Subcommittee of the International Headache Society (2004) described persistent idiopathic facial pain (PIFP) as facial pain that is present daily and persists for most of the day. Pain is confined at onset to a limited area on one side of the face, deep ache, and poorly localized. In addition, the pain is not associated with sensory loss or other physical signs, with no abnormalities in a laboratory or imaging studies.

Psychiatric symptoms of depression and anxiety are prevalent in this population. Treatment is very difficult and requires a multidisciplinary approach to address the many facets of this pain syndrome.

Treating Pain: According to BAP guidelines on the use of antidepressants (Cleare *et al.*, 2015), “There is a lack of compelling evidence that SNRIs are more effective than SSRIs for painful symptoms associated with depression. No clinically useful predictive biological factors have been identified for an antidepressant treatment response.”

D. Psychiatric aspects of HIV infection

Neuropsychiatric disorders in HIV: These can be either **primary organic** (directly due to viral CNS damage) e.g. HIV dementia or **secondary organic** (due to opportunistic infections or drugs) e.g. cerebral toxoplasmosis or **reactive** e.g. HIV-associated acute stress reaction. Prevalence of any mental disorder in the lifetime of HIV-positive patients is 38 to 73%.

1. HIV-associated acute stress reaction: This usually occurs when informing someone of seropositivity or when a transition to full clinical AIDS occurs from the state of infection. The symptoms include brooding related to their future, panic attacks, social isolation, rage or feelings of desperation. These appear within a few minutes to a few hours after the patient is informed, and remit within 2 or 3 days.

2. Adjustment Disorder: Adjustment disorder with anxiety or depressed mood has been reported to occur in 5 to 20 percent of patients infected with HIV.

3. Anxiety: Anxiety symptoms and disorders are present in 11% to 25% of HIV-infected patients. Major issues in patients infected with HIV involve self blame, self-esteem, and issues regarding death. **Worried Well** refers to those in high-risk groups who, although they are seronegative and disease free, are anxious about contracting the virus. Some are reassured by repeated negative serum test results, but others cannot be reassured. Their worried well status can progress quickly to generalized anxiety disorder, panic attacks, OCD and hypochondriasis.

4. Depression: Depression is the most common psychiatric disorders among HIV-infected. Nearly 40% (30 to 60% lifetime prevalence) of those infected with HIV has been reported to be depressed. In fact, following the introduction of HAART therapy, depression seems to be the most common cause of psychiatric hospitalisation among HIV-positive patients in some countries. Depression is higher in women than in men.

The reasons for high incidence of depression include both psychological impact and neuropathological deficits. In addition, those with high risk for HIV infection are also at a high risk for depression independent of HIV infection e.g. those with STD, homosexuals, lower socioeconomic groups, etc. In many cases, the mood disorder precedes the diagnosis of HIV infection. Furthermore, some of the criteria for depression e.g. poor sleep and weight loss can also be caused by the HIV infection itself, increasing rates of depression spuriously.

The **risk factors** for depression are similar to the risk factors for those without HIV. Poor adherence to medical treatment for HIV can occur as a result of depression, complicating management.

Suicidal ideation and suicide attempts may increase in patients with HIV infection and AIDS. The risk factors for suicide among persons infected with HIV are having friends who died from AIDS, recent notification of HIV seropositivity, relapses, difficult social issues relating to homosexuality, inadequate social and financial support, and the presence of dementia or delirium.

Antidepressants are the treatment of choice for major depression; Both TCAs and SSRIs are equally effective treatments. SSRIs are considered the safest, particularly in the case of overdose, and they tend to

have fewer side effects. SSRIs can also be used in 25 percent of the usual recommended dosage when starting.

If using a TCA for depression in HIV, most clinicians prefer secondary amines (e.g., desipramine, nortriptyline) and reserve their use for patients with the asymptomatic illness.

Patients with AIDS can respond to lower dosages of tricyclics (25–100 mg), but they may also suffer severe anticholinergic effects at reduced dosages.

Interpersonal psychotherapy may be particularly beneficial for HIV patients with depressive symptoms. Markowitz and colleagues suggested that combining psychotherapy with medication may be the optimal approach to treating depression in HIV.

5. Psychotic Disorder: Psychotic symptoms are usually later stage complications of HIV infection and are not very common. Their aetiology could well be unrelated to HIV. In some cases, it may be due to direct neurotoxicity while in others it may be iatrogenic or secondary to substance misuse. They require immediate medical and neurological evaluation and often require management with antipsychotic medications. Patients with AIDS, when treated with typical antipsychotic drugs are particularly prone to develop extrapyramidal side-effects and neuroleptic malignant syndrome. Even patients who had to stop standard neuroleptics due to extrapyramidal side effects may tolerate Risperidone.

6. Mania: HIV seems to increase the risk of manic episodes, and mania is the most frequent reason for psychiatric hospitalization among people with HIV, followed by depression and psychosis. Other associated causes include illicit drug use, iatrogenic effects (especially didanosine (ddI), ganciclovir, procarbazine, stavudine (d4T), steroids, and zidovudine (AZT)). Most cases occur in advanced HIV disease. This is often associated with the presence of substantial cognitive impairment. New-onset mania in severe symptomatic disease is predictive of reduced survival.

Development of severe adverse effects of drugs is common in manic patients with HIV. Lithium and valproate can induce neurological reactions and toxicity. The administration of carbamazepine should include strict control of these patients' haemopoietic function, above all because they are frequently taking other medications, such as AZT, which can also trigger toxic effects in the bone marrow.

7. Delirium: Delirium is one of the organic mental disorders observed most frequently in hospitalized HIV-infected patients. It is important to rule out acute causes on top of HIV infection as aetiology for delirium. Delirium in HIV infected is probably underdiagnosed.

8. AIDS Dementia: The cognitive, motor and behavioural disturbances seen in patients with AIDS were named 'AIDS dementia complex'. In 1990, WHO differentiated 'HIV-associated dementia', from 'mild cognitive/motor disorder' where ADLs are intact largely (stages 0.5 and 1 on the scale of Price) was proposed for those conditions. Dementia is noted concurrently with initial AIDS-defining illness in 3% of cases. The incidence of the dementia syndrome during the first 2 years after the diagnosis of AIDS was 7% per year. 15% of patients develop dementia during the course of the disease.

Several **risk factors** for the development of HIV-associated dementia have been reported, including older age, decreased body mass, decrements in haematocrit, and a history of intravenous drug abuse. Prognosis varies with the extent of systemic disease.

Clinically, it has an insidious onset. **Early** symptoms include

1. **Cognitive:** forgetfulness, loss of concentration, mental slowing, and reduced performance on sequential mental activities
2. Apathy, reduced spontaneity and emotional responsivity, and social withdrawal
3. **Emotional:** Depression, irritability or emotional lability, agitation, and psychotic symptoms may also occur.
4. **Motor:** loss of balance and co-ordination, clumsiness, and leg weakness;
5. In this early stage, cognitive tests show only slowing in verbal or motor responses and/or difficulty in recalling a series of objects after 5 min or more.
6. Postural tremor, hyperreflexia (particularly of the lower extremities), ataxia (usually seen only on rapid turns or tandem gait), slowing of rapid alternating movements (of the fingers, wrists or feet), frontal release signs (snout reflex, palmar grasp), dysarthria.
7. Tests of ocular motility may show an interruption of smooth pursuits, and slowing or inaccuracy of saccades.

In the **late** stages

1. Global deterioration of cognitive functions, severe psychomotor retardation. Speech can show progression to mutism.
2. Paraparesis, bladder and bowel incontinence myoclonus and seizures may occur.
3. Pedal paraesthesias and hypersensitivity may appear, due to concurrent sensory neuropathy.

Investigations: B2-microglobulin and neopterin levels in CSF and CD41 cell counts have been found to be predictive markers for the development of HIVD. A number of indirect factors may contribute to the pathogenesis of HIVD, including cytokines (e.g., TNF), parts of HIV itself (gp41, gp120, Tat, Rev, Nef), and excitatory aminoacids (e.g., quinolinic acid).

- **Neuropsychological tests** may be useful in the differential diagnosis of a depressive syndrome. Naming and vocabulary skills are largely preserved even in the most advanced cases. This pattern has been regarded as consistent with the clinical picture of a 'subcortical dementia'.
- **Brain imaging might reveal** cerebral atrophy- widened cortical sulci and, less commonly, enlarged ventricles. High-intensity T2 signal abnormalities (diffuse, patchy, focal or punctuate) most commonly located in the periventricular white matter and the centrum semiovale (less frequently, in the basal ganglia or in the thalamus). NOTE: The presence of CT / MRI finding of multiple bilateral ring-enhancing lesions suggests cerebral toxoplasmosis. Contrast-enhancing mass lesions indicate primary central nervous system lymphoma.

- **Cerebrospinal fluid analysis can help to** exclude cryptococcal meningitis; also central nervous system tuberculosis, CMV encephalitis, and neurosyphilis. In HIV-associated dementia CSF shows an increase of total proteins, the IgG fraction and mononuclear pleocytosis index. The presence of the HIV core antigen p24 or HIV RNA (PCR) can be detected even in neurologically normal subjects. HIV RNA in CSF correlates with the severity of dementia. CSF neopterin, β 2-microglobulin and quinolinic acid, interleukin 1 α , interleukin 6, tumour necrosis factor- α – are increased due to HIV invasion of CNS or even in opportunistic infections. CSF Indian ink staining, cryptococcal antigen titres and fungal culture can be decisive for the identification of cryptococcal meningitis.

Treatment: Zidovudine is a reverse transcriptase inhibitor used for treating HIV infection. Zidovudine can also have some effect on prevention of cognitive impairment. Recommendations for other medications cannot be made secondary to lack of data. (*The Annals of Pharmacotherapy*: Vol. 31, No. 4, pp. 457-473)

Didanosine does not seem to have an impact on the progression of HIV-associated dementia, whereas no studies are available on the efficacy of zalcitabine, stavudine, or lamivudine. No trial has tested the efficacy of *protease inhibitors*; according to the currently available evidence, these drugs do not reach effective concentrations in the cerebrospinal fluid. The psychostimulant *methylphenidate* (at doses ranging from 10 to 90mg/day) has been repeatedly found to improve cognitive symptoms in patients with AIDS, producing relatively mild side-effects.

9. Secondary organic brain diseases:

Progressive multifocal leucoencephalopathy is a papovavirus infection affecting white matter diffusely. Dementia can develop rapidly, with focal neurological alterations such as blindness, ataxia, and hemiparesis. Death follows very quickly thereafter, and there is no known treatment.

Cerebral toxoplasmosis involves the reactivation of a latent cerebral infection by *Toxoplasma gondii*, an opportunistic intracellular protozoan. Rapid development of a marked alteration in the mental state is noted. The lesions tend to be located in basal ganglia. Diagnosis is based on structural neuroimaging tests (multiple ring-like calcified lesions), and treatment is with pyrimethamine and sulphadiazine.

Cryptococcal meningitis (torulosis) is caused by yeast-like fungus *Cryptococcus neoformans*, It is characterized by headache, meningism, photophobia, nausea, fever, and delirium. Treatment is amphotericin B given intravenously; fluconazole and 5 flucytosine have also been tried in various combinations.

Primary CNS lymphomas (PCNSL) are a late complication of HIV infection that occur in up to 10 % of AIDS patients. The incidence of PCNSL seems to have decreased significantly. PCNSL are EBV-associated in almost 100 % of cases. Epileptic seizures, personality changes, changes in attention, headache and focal deficits without fever are the common symptoms.

10. Antiretroviral treatment –psychiatric effects:

Antiretroviral	Effects
Zidovudine	Confusion, agitation, insomnia, mania and depression reported.
Stavudine, didanosine and zalcitabine	Can cause peripheral neuropathy. Mania is associated with didanosine
Efavirenz	Significant psychiatric side effects known. 46% of patients developed neuropsychiatric side effects (1/3 rd had depression, 2% psychosis) while on this drug while nearly 6% may stop the drug due to these effects.

DISCLAIMER: This material is developed from various revision notes assembled while preparing for MRCPsych exams. The content is periodically updated with excerpts from various published sources including peer-reviewed journals, websites, patient information leaflets and books. These sources are cited and acknowledged wherever possible; due to the structure of this material, acknowledgements have not been possible for every passage/fact that is common knowledge in psychiatry. We do not check the accuracy of drug-related information using external sources; no part of these notes should be used as prescribing information

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